

Notes: Ivashina, Scharfstein, and Stein (QJE, 2015)

Dollar Funding and the Lending Behavior of Global Banks

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1 Big picture

- **Question.** How do shocks to global banks' dollar funding affect their lending behavior? The paper asks whether foreign banks that rely on wholesale dollar funding cut dollar lending more than local-currency lending when their credit quality deteriorates.
- **Setting.** The empirical setting is the Eurozone sovereign crisis, especially May 2011 to June 2012. During this period, U.S. prime money market funds sharply reduced funding to Eurozone banks.
- **Core friction.** Eurozone banks finance much of their euro lending with relatively sticky insured retail deposits, but finance dollar lending with more flighty wholesale dollar liabilities, such as commercial paper and certificates of deposit held by U.S. money market funds.
- **Key financial-market mechanism.** Banks could in principle borrow euros and swap them into dollars. But if arbitrage capital in the FX swap market is limited, increased demand for dollar swaps creates covered interest parity (CIP) deviations. Synthetic dollar borrowing becomes expensive too.
- **Main prediction.** A negative shock to bank credit quality should reduce dollar lending more than euro lending. The effect should be strongest for banks most reliant on U.S. money market funds.
- **Main result.** During the Eurozone funding shock, Eurozone banks reduced the dollar share of their syndicated lending, were less likely to appear in dollar-loan syndicates, and their relationship borrowers found it harder and more expensive to borrow in dollars.
- **Economic takeaway.** The global role of the dollar creates a funding fragility: foreign banks can transmit shocks from their own balance sheets into dollar credit markets, even for borrowers outside the banks' home countries.

2 Incremental contribution of the paper

- **From bank capital shocks to currency-specific funding shocks.** Earlier work studies how weak banks cut lending overall. This paper shows that banks may cut lending differentially by currency because funding sources differ by currency.
- **A model linking wholesale dollar funding, CIP deviations, and real lending.** The paper connects the breakdown of covered interest parity to banks' loan supply decisions. CIP deviations are not just a financial-market anomaly; they can affect the real allocation of credit.
- **Evidence on non-U.S. banks in dollar credit markets.** The paper documents that foreign banks, especially European banks, are major suppliers of dollar-denominated credit, including in the U.S. syndicated loan market.
- **Clean event-window evidence from the Eurozone crisis.** The Eurozone sovereign shock created a plausibly sharp deterioration in perceived Eurozone bank credit quality, followed by a pullback of U.S. money market fund funding.
- **Borrower fixed-effect design.** By comparing repeated loans to the same borrower, the paper reduces concern that the result is driven by dollar-loan demand rather than Eurozone-bank loan supply.

- **Real effects on borrowers.** The paper goes beyond bank-level loan shares and shows that firms previously dependent on Eurozone banks were less likely to obtain dollar loans and paid higher spreads if they borrowed.
- **Policy relevance.** The analysis gives a framework for understanding dollar swap lines, liquidity regulation of foreign banking organizations, and the systemic implications of the dollar’s global funding role.

3 Data and measurement

3.1 Syndicated loan data

- The loan data come from Thomson Reuters DealScan and cover syndicated loan originations.
- Loans are assigned to lead banks using the “lead arranger” or “agent” designations. When there are multiple lead banks, the loan amount is prorated across them.
- The paper distinguishes Eurozone banks, non-Eurozone European banks, U.S. banks, and other banks using bank headquarters.
- The key lending currencies are U.S. dollars and euros. Other currencies are excluded from the main dollar-share regressions.

Interpretation: DealScan allows the authors to observe the currency of lending, the identity of lenders, the identity of borrowers, loan sizes, and loan spreads. This makes it possible to separate a bank’s dollar lending from its euro lending.

3.2 Money market fund funding data

- U.S. money market fund holdings are used to measure Eurozone banks’ reliance on short-term dollar funding.
- The main bank-level measure, *MMF reliance*, equals U.S. money market fund holdings of a bank’s short-term instruments divided by the bank’s deposits plus short-term debt.
- The paper emphasizes that this denominator includes all short-term funding, not only dollar short-term funding. The measure therefore understates reliance on MMFs as a share of dollar funding.
- The paper classifies banks as more or less MMF exposed using the pre-shock April 2011 reliance measure.

Interpretation: The MMF data identify which Eurozone banks should be most exposed to the withdrawal of U.S. wholesale dollar funding.

3.3 Covered interest parity basis

- The paper measures deviations from CIP using three-month sovereign bill rates and spot/forward exchange rates.

- A positive dollar-euro basis indicates that synthetic dollar borrowing through FX swaps is expensive relative to the textbook CIP benchmark.
- During the shock period, the dollar-euro basis widened substantially, consistent with high demand for synthetic dollar borrowing by European banks and limited arbitrage capital.

Interpretation: The CIP basis is the market price of the friction in the model. When the basis rises, swapping euro funding into dollar funding is no longer a frictionless substitute for direct dollar funding.

3.4 Borrower outcomes

- The paper studies whether a borrower obtains a dollar loan during the shock period.
- It measures the borrower's pre-shock dependence on Eurozone banks using the Eurozone-bank share of the borrower's last pre-shock dollar-loan syndicate.
- For borrowers that do obtain new dollar loans, the paper studies the change in the all-in-drawn spread over LIBOR relative to the borrower's last pre-shock loan of the same facility type.

Interpretation: These borrower-level outcomes test whether the Eurozone-bank funding shock was absorbed by other lenders or instead reduced the overall availability of dollar credit.

4 Models and empirical specifications

4.1 Model intuition

The paper considers a global bank that lends in dollars and euros. Dollar lending is denoted by L_D and euro lending by L_E . The bank has a binding capital constraint:

$$L_D + L_E \leq K.$$

Dollar lending earns expected gross return $g(L_D)$ and euro lending earns expected gross return $h(L_E)$, with both functions concave.

The bank's probability of default is p . Its euro borrowing is protected by deposit insurance and therefore has a relatively sticky funding cost. Its dollar borrowing relies more on uninsured wholesale funding and has a cost that rises with credit risk, approximately:

$$r_D^B = r + \alpha p.$$

The bank can borrow euros and swap into dollars, but this requires using FX swaps. If arbitrageurs have limited capital, swap demand creates a CIP basis. In an interior equilibrium, synthetic dollar funding is also costly. The model implies:

$$g'(L_D) = h'(L_E) + \text{CIP basis}.$$

Interpretation: When the bank becomes riskier, its direct dollar funding becomes more expensive. If the swap market cannot absorb all demand frictionlessly, synthetic dollar borrowing also becomes expensive. The bank reallocates lending away from dollars and toward euros.

4.2 Bank-month dollar lending share

The first empirical object is a bank-month dollar loan share:

$$\text{DollarShare}_{it} = \frac{\text{Dollar loans}_{it}}{\text{Dollar loans}_{it} + \text{Euro loans}_{it}}.$$

The Eurozone-bank specification is:

$$\text{DollarShare}_{it} = \alpha_i + \beta \text{Shock}_t + \Gamma F X_t + \varepsilon_{it},$$

where Shock_t equals one between May 2011 and June 2012.

The difference-in-difference specification adds U.S. banks:

$$\text{DollarShare}_{it} = \alpha_i + \delta_t + \beta(\text{Eurobank}_i \times \text{Shock}_t) + \Gamma F X_t + \varepsilon_{it}.$$

Interpretation: If the coefficient is negative for Eurozone banks but not for U.S. banks, the pattern is more consistent with a Eurozone-bank dollar funding shock than with a general fall in demand for dollar loans.

4.3 Loan-level lender composition

At the loan level, the paper uses the share of lead banks in the syndicate headquartered in the Eurozone:

$$\text{EurobankShare}_{jt} = \alpha_j + \beta \text{Shock}_t + X_j' \Gamma + \varepsilon_{jt},$$

for dollar loans, where α_j are borrower fixed effects.

A richer version includes both dollar and euro loans:

$$\text{EurobankShare}_{jt} = \alpha_j + \delta_t + \beta(\text{DollarLoan}_{jt} \times \text{Shock}_t) + X_j' \Gamma + \varepsilon_{jt}.$$

Interpretation: The borrower fixed effects compare the same firm across time. The interaction asks whether Eurozone banks specifically retreat from dollar lending, not from all lending.

4.4 Borrower-level real effects

The paper then asks whether borrowers exposed to Eurozone banks before the shock can still borrow in dollars during the shock. The key variable is:

$$\text{PastEurobankShare}_j = \text{extfractionofEurozoneleadbanksinthelastpre} - \text{shockdollarloan}.$$

The extensive-margin outcome is an indicator for obtaining a dollar loan during May 2011-June 2012. The intensive-margin outcome is the change in loan spread for borrowers that obtain a new loan.

Interpretation: If other lenders fully replace Eurozone banks, past Eurozone-bank dependence should not matter. The paper finds that it does matter.

4.5 Heterogeneity by MMF reliance

The model predicts stronger dollar lending cuts for banks more dependent on flighty dollar funding. The paper estimates the Table V loan-level design separately for Eurozone banks with high versus low money fund reliance.

Interpretation: This is a cross-sectional mechanism test. The banks most exposed to the dollar funding run should retreat the most from dollar lending.

5 Identification and empirical design

5.1 What identifies the shock

- The shock period corresponds to heightened Eurozone sovereign stress and the withdrawal of U.S. MMF funding from Eurozone banks.
- The timing is economically meaningful: concerns about Greek debt and Eurozone bank credit quality intensified in 2011, and U.S. MMFs sharply reduced exposures.
- The shock affects Eurozone banks' dollar funding more directly than it affects U.S. banks or Eurozone banks' euro retail deposit funding.

Interpretation: The event is not a randomized experiment, but it is a sharp funding shock that maps closely to the model's mechanism.

5.2 Controls for loan demand

- The paper compares dollar and euro lending by the same Eurozone banks.
- It compares Eurozone banks with U.S. banks.
- It restricts some tests to the European market to reduce the concern that U.S. and European borrowers have different demand shocks.
- It uses borrower fixed effects in loan-level regressions, comparing repeated loans to the same firm.

Interpretation: The strongest demand-side control is the borrower fixed-effect design: the same borrower is less likely to have Eurozone banks in dollar-loan syndicates during the shock.

5.3 What the design can and cannot prove

- The design is strongest for showing that Eurozone banks reduced dollar lending relative to euro lending during a period when they lost dollar funding.
- It is also strong for showing that borrowers with prior Eurozone-bank exposure had worse dollar borrowing outcomes.
- It cannot perfectly separate every possible supply-side channel. For example, capital stress and dollar funding stress may move together.

- The cross-sectional MMF reliance result helps isolate the specific dollar-funding mechanism.

Interpretation: The paper’s causal argument is a triangulation: model prediction, funding event, CIP basis movement, bank-level lending shifts, borrower fixed effects, borrower real effects, and MMF-exposure heterogeneity all point in the same direction.

6 Tables and figures

6.1 Figures I and II: global dollar balance sheets and the MMF pullback

Read:

- Figure I shows that foreign banks’ dollar assets and dollar liabilities rose dramatically from the 1990s to the 2000s, reaching roughly \$10 trillion.
- The figure motivates why non-U.S. banks matter for dollar credit markets.
- Figure II shows a sharp reduction in MMF exposure to European and Eurozone banks during the shaded May 2011-June 2012 period.
- Eurozone-bank exposure falls especially sharply, consistent with a run-like withdrawal of dollar wholesale funding.

Economic message: Foreign banks are central to dollar intermediation, but their dollar funding can disappear quickly. The figures jointly motivate the paper’s central fragility: a global dollar lending market funded by flighty wholesale dollar liabilities.

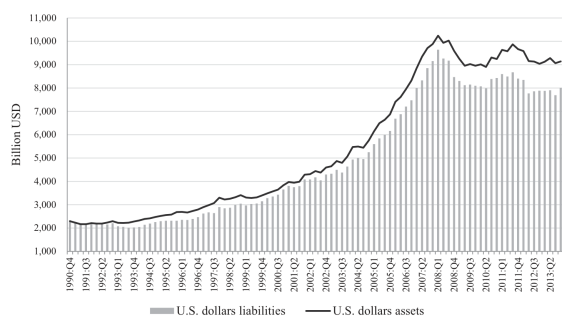


FIGURE I

Dollar Assets and Liabilities of Foreign Banks

Compiled from Bank of International Settlement locational banking statistics, July 2014.

(a) Figure I: foreign-bank dollar assets and liabilities

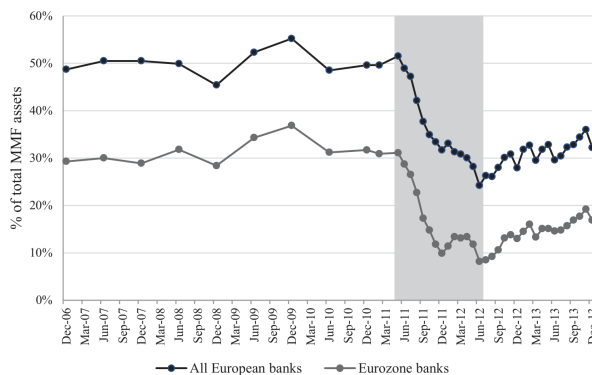


FIGURE II

Money Market Fund Exposure to European Banks

The figure shows the fraction of money market fund assets invested in liabilities of European and Eurozone banks. Data are from Fitch Ratings, “U.S. Money Fund Exposure and European Banks,” February 4, 2014. The data are monthly starting in February 2011, semiannual before that. The high-lighted area corresponds to the period May 2011 through June 2012. (a) Euro basis, January 2007–December 2013. (b) Other currencies, July 2010–December 2013

(b) Figure II: MMF exposure to European banks

6.2 Figures III and IV: CIP deviations and lender composition

Read:

- Figure III shows that the dollar-euro CIP basis widened during the Eurozone funding shock, indicating that synthetic dollar borrowing through FX swaps became costly.
- The basis also widens for other currencies, consistent with broader constraints on arbitrage capital.
- Figure IV plots the Eurozone-bank share in dollar-loan syndicates after controlling for borrower fixed effects and loan size.
- The Eurozone-bank share is lower in the May 2011-June 2012 shock period than in surrounding periods.

Economic message: Figure III shows the price of the funding friction; Figure IV shows the lending response. Together they connect swap-market stress to real dollar loan supply.

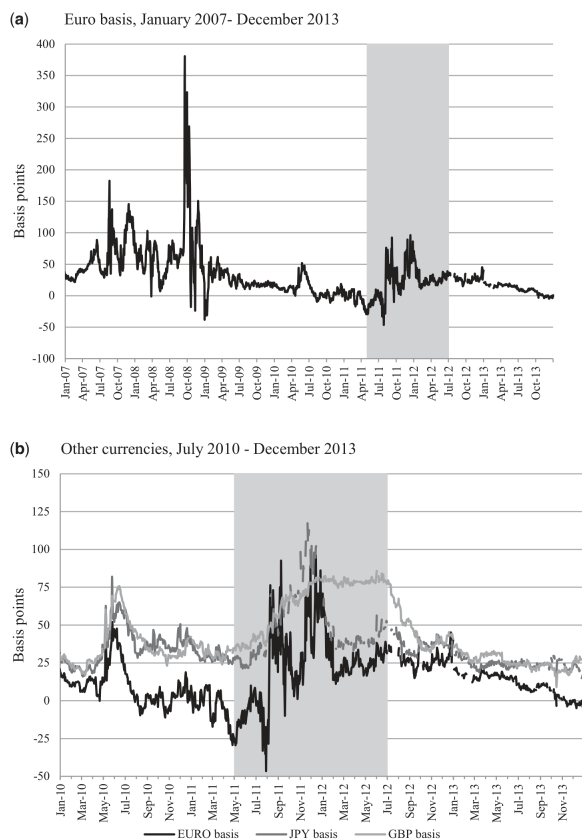


FIGURE III

Deviations from Covered Interest Parity

The basis is constructed using sovereign bonds rates. Daily quotes for three-month sovereign rates are from Bloomberg, and three-month daily spot and forward rates are compiled by WM/Reuters and downloaded from Datastream. The highlighted area corresponds to May 2011 through June 2012.

(a) Figure III: deviations from covered interest parity

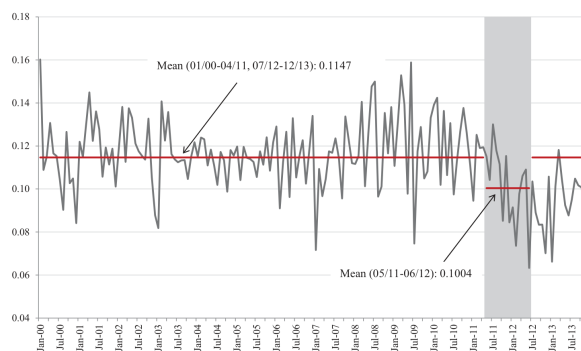


FIGURE IV

Choice of Lender for Dollar-Denominated Loans

This figure illustrates graphically the source of identification in the regressions reported in Table V. The plotted series corresponds to the coefficients (β_i) on the monthly dummies of the following specification: $EUROBANK SHARE_{it} = D_j + \beta_i D_t + X_i$, where the D_j are borrower fixed effects, the D_t are month fixed effects, and X_i is loan size. The sample includes U.S. dollar denominated loans issued in Europe and U.S. The highlighted area corresponds to May 2011 through June 2012. The difference between average β_i for this period and the rest of the sample ($0.1147 - 0.1004 = -0.0143$) is equivalent to the coefficient on *SHOCK* in specification (2) in Table V. However, the magnitudes are not exactly the same because the panel is unbalanced.

(b) Figure IV: choice of lender for dollar loans

6.3 Table I: market shares of major lenders

Read:

- Table I reports the largest lenders in the Eurozone and U.S. syndicated loan markets in 2005-2007.
- Eurozone banks account for a meaningful share of U.S. syndicated lending, with large French and German banks especially visible.
- Total European banks account for a substantially larger U.S. market share than Eurozone banks alone, but the paper focuses conservatively on Eurozone banks.
- The table shows why a Eurozone-bank dollar funding shock can matter for firms borrowing in U.S. dollars.

Economic message: Eurozone banks are not marginal lenders. They are large enough in dollar syndicated lending that their funding shocks can affect aggregate credit supply and borrower outcomes.

TABLE I
MARKET SHARE, LARGEST LENDERS IN EUROZONE, 2005-2007

	Market	Share of Eurozone lending		Market share U.S.	
		Lead	All lenders	Lead	All lenders
Europe, "GIIPS"					
UniCredit	Italy	3.02	3.21	0.05	0.31
Banco Bilbao Vizcaya Argentaria	Spain	2.33	2.23	0.06	0.43
Intesa Sanpaolo	Italy	1.91	2.14	0.04	0.26
Banco Santander	Spain	2.37	2.32	0.10	0.44
Mediobanca	Italy	0.99	1.07	—	—
Banco Financiero y de Ahorros	Spain	0.90	1.00	0.01	0.04
La Caja de Barcelona	Spain	0.78	0.77	—	—
Banco de Sabadell	Spain	0.32	0.35	—	—
Bank of Ireland Group	Ireland	0.30	0.45	0.05	0.24
ICO [Instituto de Credito Oficial]	Spain	0.23	0.26	—	—
France					
BNP Paribas	France	6.70	5.07	2.31	2.36
Credit Agricole	France	4.91	4.29	1.03	1.25
Societe Generale	France	4.57	3.64	0.48	1.02
Natixis SA	France	3.01	3.22	0.09	0.47
CM-CIC	France	1.32	1.72	0.00	0.06
Benelux					
ING Group	Netherlands	2.72	2.47	0.33	0.81
Fortis Bank	Belgium	1.71	1.89	0.38	0.66
Rabobank	Netherlands	0.78	0.96	0.30	0.41
KBC Group	Belgium	0.47	0.74	0.02	0.28
Dexia Bank	Belgium	0.46	0.65	0.02	0.06
Rest of Eurozone					
Commerzbank	Germany	4.92	4.74	0.50	0.90
Deutsche Bank	Germany	4.74	3.50	4.47	3.25
WestLB	Germany	1.32	1.41	0.14	0.29
BayernLB	Germany	1.13	1.22	0.07	0.23
Landesbank	Germany	0.98	1.14	0.01	0.04
Baden-Wuerttemberg Landesbank	Germany	0.42	0.62	0.04	0.08
Hessen-Thuringen [Helaba]	Germany	0.42	0.56	0.01	0.10
DZ Bank	Germany	0.41	0.68	0.00	0.01
KfW Bankengruppe	Germany	0.38	0.38	0.06	0.09
HSH Nordbank	Germany	0.24	0.31	0.03	0.08
NordLB Group	Germany	0.24	0.31	0.03	0.08
Total Eurozone		54.77	53.00	10.61	14.15

Continued

TABLE I
(CONTINUED)

	Market	Share of Eurozone lending		Market share U.S.	
		Lead	All lenders	Lead	All lenders
Rest of Europe					
Svenska Handelsbanken	Sweden	0.31	0.26	—	—
SEB Merchant Banking	Sweden	0.29	0.40	—	0.04
Nordea Bank AB	Sweden	0.45	0.46	0.03	0.09
Credit Suisse	Switzerland	1.60	1.64	3.76	0.09
UBS	Switzerland	1.26	1.23	1.73	3.37
Royal Bank of Scotland	UK	9.21	7.71	3.87	2.01
Barclays Bank	UK	3.16	2.87	2.27	3.77
HSBC	UK	3.08	2.87	1.18	2.07
Lloyds Banking Group	UK	1.01	1.33	0.39	1.57
Total Europe		75.14	71.75	23.86	27.68
United States					
Citi	USA	5.66	4.03	11.45	6.70
JP Morgan	USA	3.75	3.13	17.08	10.13
Bank of America	USA	1.86	1.94	16.15	9.83
Merrill Lynch					
Goldman Sachs	USA	1.68	1.79	2.55	2.71
Morgan Stanley	USA	1.32	1.43	1.68	2.16
Lehman Brothers	USA	0.52	0.60	2.26	2.18
GE Capital	USA	0.52	0.56	2.36	2.29
Total United States		15.29	13.49	53.53	36.00
Other					
Mitsubishi UFJ	Japan	2.27	2.47	1.85	2.45
Financial Group					
Mizuho Financial Group	Japan	0.96	1.08	0.56	1.35
Sumitomo Mitsui	Japan	0.41	0.61	0.37	0.98
Financial Group					
Total other		3.64	4.15	2.78	4.77

Notes. Loan amount is prorated based on the number of the lead banks ("Lead") or based on the total number of syndicate participants ("All lenders"). Lead bank is identified based on whether the lender is designated as "Lead Arranger" or "Agent" in the league tables as reported in DealScan. The table reports the top 50 lenders in the Eurozone; Banesto's share is aggregated together with Santander's share. Note that Lehman Brothers went bankrupt in 2008, and Fortis was acquired by BNP Paribas in 2009.

(a) Table I, part 1

(b) Table I, part 2

6.4 Tables II and III: dollar lending and MMF funding reliance

Read:

- Table II shows that Eurozone banks lend globally and that a large share of their syndicated lending is dollar-denominated.
- Outside Europe, Eurozone banks do most syndicated lending in dollars; even within Europe, a nontrivial share is in dollars.
- Table III reports pre-shock MMF reliance and the subsequent decline in MMF funding for major European banks.
- French banks and some other Eurozone banks lose especially large shares of their MMF funding during the shock.

Economic message: Table II establishes the asset side of the mismatch: Eurozone banks hold large dollar loan books. Table III establishes the liability-side fragility: some of those banks rely heavily on U.S. MMFs for short-term funding.

TABLE II
SYNDICATED LENDING AROUND THE WORLD BY CURRENCY, 2005-2007

	Lending in the region	% of total lending	Lending in Euro (US\$billion)	Lending in USD (US\$billion)	Lending in euro (%)	Lending in USD (%)
Eurozone banks						
Eurozone	1,036.15	45.5	941.64	89.11	90.9	8.6
Rest of Europe	420.64	18.5	124.04	127.12	29.5	30.2
United States	495.52	21.8	4.30	490.53	0.9	99.0
Rest of North America	17.89	0.8	0.00	14.47	0.0	80.9
Asia	139.44	6.1	10.06	102.92	7.2	73.8
Latin America	81.09	3.6	2.27	78.81	2.8	97.2
Middle East	63.25	2.8	3.40	59.78	5.4	94.5
Africa	21.27	0.9	1.74	17.65	8.2	83.0
Total	2,275.25		1,087.46	980.39	47.8	43.1
U.S. banks						
Eurozone	313.96	7.8	252.06	60.53	80.3	19.3
Rest of Europe	234.74	5.8	41.48	75.46	17.7	32.1
U.S.	3,269.47	80.8	7.82	3,255.99	0.2	99.6
Rest of North America	56.83	1.4	0.12	47.79	0.2	84.1
Asia	89.91	2.2	5.56	60.11	6.2	68.9
Latin America	56.79	1.4	0.41	56.38	0.7	99.3
Middle East	15.54	0.4	0.16	15.38	1.0	99.0
Africa	10.24	0.3	0.32	8.69	3.1	84.9
Total	4,047.48		307.93	3,580.33	7.6	88.5

Note: Loan amount is presented based on the number lead banks receiving loans table credit.

(a) Table II: syndicated lending by region and currency

TABLE III
U.S. PRIME MONEY MARKET FUNDS AS A FUNDING SOURCE

Bank	Country	MMF reliance as of April 2011 (%)	Change in MMF reliance, April 2011-June 2012 (%)
Eurozone			
Deutsche Bank	Germany	7.65	2.4
Rabobank	Netherlands	7.63	-2.7
Societe Generale	France	6.25	-67.6
ING Bank	Netherlands	5.14	-27.8
Natixis	France	5.06	-100*
Credit Agricole	France	4.28	-82.3
BNP Paribas	France	4.25	-77.1
Commerzbank	Germany	1.90	-100*
Banco Bilbao	Spain	1.25	-100*
Vizcaya Argentaria			
UniCredit	Italy	0.99	-100*
Banco Santander	Spain	0.78	-100*
Rest of Europe			
Credit Suisse	Switzerland	4.92	41.9
Barclays Bank	UK	4.37	-14.8
Lloyds TSB Bank	UK	3.97	-62.3
UBS	Switzerland	3.32	-49.4
RBS	UK	2.29	-61.7
HSBC	UK	1.49	n.a.

Notes. This table reports money market fund (MMF) reliance for the 11 Eurozone banks that were among the top 50 lenders in the U.S. syndicated loan market between 2005 and 2007. We also include MMF data for those European banks outside the Eurozone that were among the top 50 lenders in the United States. *MMF reliance* equals MMF holdings as of April 2011 divided by (deposits + short-term debt) as of the end of 2010. *Change in MMF reliance* is compiled from multiple Fitch Ratings reports on U.S. MMF exposure to European banks. Fitch reports highlight banks with the largest use of MMFs by country. Some of the banks or even entire countries are dropped from the coverage when their use of MMFs becomes very small; those cases are indicated by an asterisk.

(b) Table III: U.S. prime MMFs as funding source

6.5 Tables IV and V: bank-level and loan-level lending response

Read:

- Table IV shows that Eurozone banks reduce their dollar loan share during the shock period by roughly 3.5 percentage points.
- Including U.S. banks produces a difference-in-difference result: the negative effect is specific to

Eurozone banks.

- Table V moves to the loan level and uses borrower fixed effects.
- For dollar loans, syndicates are less likely to include Eurozone banks during the shock; the effect is concentrated in dollar loans relative to euro loans.

Economic message: These are the core loan-supply tables. They show that the shock is not merely associated with lower lending overall; it changes the currency composition of Eurozone-bank lending and the composition of dollar-lending syndicates.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Banks	Eurozone banks				Eurozone and U.S. banks			
Market	Europe and U.S.		Europe		Europe and U.S.			
	Loan volume	Number of loans	Loan volume	Number of loans	Loan volume	Number of loans	Loan volume	Number of loans
<i>SHOCK</i> (05/2011–06/2012)	−0.0348*	−0.0328**	−0.0365*	−0.0342***	0.0053	—	−0.0007	—
	[0.020]	[0.013]	[0.020]	[0.013]	[0.014]	[0.002]	[0.013]	[0.014]
<i>EUROBANK*SHOCK</i>	—	—	—	—	−0.0398*	−0.0398*	−0.0313**	−0.0313**
	[0.020]	[0.020]	[0.020]	[0.020]	[0.021]	[0.021]	[0.013]	[0.014]
USD/euro spot exchange rate	0.1182	0.1535***	0.1282*	0.1577***	0.0916*	—	0.0996***	—
	[0.073]	[0.050]	[0.075]	[0.050]	[0.054]	[0.032]	[0.032]	[0.032]
Fixed effects:								
Bank (D_i)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month (D_t)	—	—	—	—	—	—	—	—
Observations	1,188	1,188	1,188	1,188	1,836	1,836	1,836	1,836
Clusters (D_i)	108	108	108	108	108	108	108	108
Adj. R -squared	0.09	0.14	0.09	0.14	0.83	0.86	0.91	0.91

Notes. The dependent variable is the fraction of loans originated by bank i in month t that is denominated in U.S. dollars (S_{it}). The sample includes all loans originated between 2005 and 2013 that are denominated in U.S. dollars or euros; all other currencies are excluded from the sample. In specifications (1)–(4) we look at Eurozone banks only; specifications (5)–(8) look at U.S. and Eurozone banks. Specifications (2) and (4) look at lending in the European market only; the rest of the specifications look at lending in U.S. and European markets. Specifications (1)–(4) correspond to $S_{it} = D_i + \beta SHOCK + \epsilon_{it}$, D_i is an originating bank fixed effect and *SHOCK* is a dummy variable equal to 1 for the May 2011–June 2012 period and 0 otherwise. Specification (5) includes month fixed effects and corresponds to $S_{it} = D_i + D_t + \beta SHOCK + \epsilon_{it}$. *EUROBANK* is a dummy variable equal to 1 if bank's headquarters are located in the Eurozone and 0 otherwise. Standard errors, reported in brackets, are clustered by month. The average of the dependent variable (S_{it}) is 84.5 percent for Eurozone banks and 85.3 percent for U.S. banks; thus the high R -squared in specifications (5)–(8) is due to increased explanatory power of bank fixed effects. Significance at the 1 percent, 5 percent, and 10 percent levels is indicated by ***, **, and *, respectively.

(a) Table IV: share of dollar lending

	(1)	(2)	(3)	(4)	(5)
Market	U.S. dollar		Europe and U.S.		
Loan currency	U.S. dollar		U.S. dollar and euro		
<i>SHOCK</i>	−0.0105***	−0.0119**	0.0682***	—	—
	[0.005]	[0.005]	[0.019]	[0.019]	[0.023]
<i>DOLLAR LOAN*SHOCK</i>	—	—	−0.0763***	−0.0758***	−0.0762***
	[0.019]	[0.019]	[0.019]	[0.019]	[0.023]
<i>DOLLAR LOAN</i>	—	—	−0.0673***	−0.0664***	−0.0664***
	[0.022]	[0.022]	[0.022]	[0.022]	[0.022]
Ln(loan amount)	—	0.0041**	−0.0017	−0.0021	−0.0011
	[0.002]	[0.002]	[0.002]	[0.002]	[0.001]
Fixed effects:					
Borrower (D_i)	Yes	Yes	Yes	Yes	—
Month (D_t)	—	—	—	—	Yes
Industry (D_{SIC})	—	—	—	—	Yes
Observations	32,309	32,303	37,236	37,236	36,940
Clusters (D_i)	168	168	168	168	168
Adj. R -squared	0.73	0.73	0.83	0.83	0.42

Notes. Each observation used for the analysis reported in this table is a separate loan. The dependent variable is *EUROBANK SHARE*, a variable between 0 and 1 equal to fraction of loan banks on the loan headquartered in the Eurozone. Specifications (1) and (2) include only U.S. dollar-denominated loans and correspond to *EUROBANK SHARE* = $D_i + \beta SHOCK + \epsilon_{it}$. The rest of specifications include U.S. dollar-denominated and euro loans. In particular, specification (4) corresponds to *EUROBANK SHARE* = $D_i + D_t + \beta DOLLAR LOAN + \beta SHOCK + \epsilon_{it}$, where the D_t are borrower fixed effects. That is, the coefficient of interest, β , is identified off repeated loans to the same borrower. The D_t are month fixed effects. *DOLLAR LOAN* is a dummy for the loan being denominated in U.S. dollars. *SHOCK* is a dummy variable equal to 1 for May 2011–June 2012 and 0 otherwise. X_i is loan size; we include it as a control because the number of loan lenders depends on loan size. We use loans issued over 2000–2013 period to ensure that there are enough repeated loans in our sample. In specification (5), we replace borrower fixed effects with two-digit Standard Industrial Classification (SIC) code fixed effects. Standard errors, reported in brackets, are clustered by month. Significance at the 1 percent, 5 percent, and 10 percent levels is indicated by ***, **, and *, respectively.

(b) Table V: choice of lender for dollar loans

6.6 Tables VI and VII: borrower real effects

Read:

- Table VI asks whether firms obtain a dollar loan during the shock period.
- Borrowers with greater pre-shock exposure to Eurozone-bank syndicates are less likely to obtain a new dollar loan during the shock.
- Table VII focuses on borrowers that do receive a new dollar loan.
- These borrowers pay higher spreads when their last pre-shock syndicate had a larger Eurozone-bank share.

Economic message: Other lenders do not fully replace Eurozone banks. Relationship borrowers face lower access to dollar credit and higher prices, implying that the bank funding shock has real borrower-level consequences.

TABLE VI
LIKELIHOOD OF OBTAINING A DOLLAR-DENOMINATED LOAN

Market	(1)	(2)	(3)	(4)
	Europe and U.S.			
Loan currency	U.S. dollar			
<i>PAST EURO BANK SHARE</i> (last loan)	-0.0542*** [0.011]	-0.0418*** [0.012]	-0.0487*** [0.012]	—
<i>PAST EURO BANK SHARE</i> (5-year avg)	—	—	—	-0.0635*** [0.012]
All-in-drawn spread	—	—	-0.0002*** [0.000]	-0.0002*** [0.000]
No spread information	—	—	-0.0601*** [0.011]	-0.0592*** [0.011]
Ln(loan amount)	—	—	0.0295*** [0.003]	0.0297*** [0.003]
Maturity	—	—	0.0038** [0.002]	0.0038** [0.002]
Revolving line	—	—	0.0310*** [0.008]	0.0303*** [0.008]
Fixed effects:				
Industry (D_{SIC})	—	Yes	Yes	Yes
Year of last loan origination	—	Yes	Yes	Yes
Observations	10,190	9,444	9,441	9,441
R-squared	0.00	0.21	0.23	0.23

Notes. The dependent variable is an indicator for whether the borrower received a U.S. dollar-denominated loan between May 2011 and June 2012 (the "shock period"). Due to the presence of fixed effects, the model is estimated as a linear probability model. The sample includes all borrowers that received a dollar-denominated loan before the shock period. With the exception of borrower two-digit SIC code fixed effects, control variables correspond to the last preshock dollar-denominated loan. The explanatory variable of interest in specifications (1)–(3) is *PAST EURO BANK SHARE*, a variable between 0 and 1 equal to fraction of lead banks on the last preshock loan headquartered in the Eurozone. In specification (4), we compute *PAST EURO BANK SHARE* as the average of the last preshock loan and any other loans issued in the preceding five years. Robust standard errors are reported in brackets. Significance at the 1 percent, 5 percent, and 10 percent levels is indicated by ***, **, and *, respectively.

(a) Table VI: likelihood of obtaining a dollar loan

TABLE VII
THE EFFECT OF LENDER TYPE ON LOAN SPREADS

Market	(1)	(2)	(3)
	Europe and U.S.		
Loan currency	U.S. dollar		
<i>PAST EURO BANK SHARE</i>	34.577* [18.085]	45.09*** [16.059]	39.073** [15.704]
All-in-drawn spread	—	—	-0.439*** [0.048]
Ln(loan amount)	—	—	-18.219*** [3.169]
Maturity	—	—	13.851*** [2.531]
Loan type	—	—	-74.047*** [9.104]
Fixed effects:			
Industry (D_{SIC})	Yes	Yes	Yes
Year of last loan origination	—	Yes	Yes
Observations	3,816	3,816	3,816
Clusters (D_j)	1,413	1,413	1,413
R-squared	0.08	0.33	0.45

Notes. The dependent variable is the increase in interest spread charged over LIBOR for U.S. dollar-denominated loans issued between May 2011 and June 2012 (the "shock period") as compared to the spread on the last dollar loan received by the same borrower before the shock. The explanatory variable of interest is *PAST EURO BANK SHARE*, a variable between 0 and 1 equal to fraction of lead banks on the last preshock loan headquartered in the Eurozone. The last two specifications also control for the year of origination of the last preshock loan. Spread, loan amount, and maturity reflect characteristics of the last loan issued before the shock period. The spread is measured at the loan facility or tranche level. The interest spread on the shock facilities are matched to the last preshock facility of the same type. Loans in a matched pair must be revolving lines, bank tranches, or institutional tranches. We exclude second-lien facilities. Standard errors are clustered by borrower. Significance at the 1 percent, 5 percent, and 10 percent levels is indicated by ***, **, and *, respectively.

(b) Table VII: loan spreads

6.7 Table VIII: money fund reliance mechanism test

Read:

- Table VIII separates Eurozone banks into those with high and low reliance on U.S. MMFs before the shock.
- The shock effect is concentrated among the most MMF-exposed Eurozone banks.
- The difference between high- and low-exposure banks is statistically significant in the reported F-test.

Economic message: This table strengthens the paper's mechanism. If the retreat from dollar lending is really caused by dollar funding pressure, then banks most exposed to MMFs should retreat more. That is what the table shows.

TABLE VIII
THE EFFECT OF MONEY FUND RELIANCE

Market	(1)	(2)
	Europe and U.S.	
	Most MMF exposed banks	Least MMF exposed banks
<i>SHOCK</i>	−0.0095** [0.004]	−0.0024 [0.002]
Ln(loan amount)	0.0032** [0.001]	0.0009 [0.001]
Fixed effects:		
Borrower (D_j)	Yes	Yes
Observations	32,303	32,303
Clusters (D_t)	168	168
Adj. R -squared	0.71	0.68

H₀: $\beta_{SHOCK, \text{most MMF exposed banks}} = \beta_{SHOCK, \text{least MMF exposed banks}}$

F -statistic = 6.20**

Notes. The specifications reported here are analogous to Table V, column (2), the difference is the dependent variable. In specification (1), the dependent variable is the fraction of lead banks on the loan that are headquartered in the Eurozone and, as of April 2011, had over 4 percent of their short-term funding coming from U.S. MMFs. In specification (2), the dependent variable is the fraction of lead banks on the loan that are headquartered in the Eurozone and, as of April 2011, had less than 4 percent of their short-term funding coming from U.S. MMFs. As before, *SHOCK* is a dummy variable equal to 1 for the May 2011–June 2012 period and 0 otherwise. We use loans issued over 2000–2013 to ensure that there are enough repeated loans in our sample. We control for loan size because the number of lead lenders depends on loan size. Standard errors, reported in brackets, are clustered by month. Significance at the 1 percent, 5 percent, and 10 percent levels is indicated by ***, **, and *, respectively.

Table VIII: effect of money fund reliance

7 Short summary

- The paper studies how global banks respond when their dollar funding becomes scarce.
- Non-U.S. banks are large players in dollar intermediation, and Eurozone banks are important lenders in the U.S. syndicated loan market.
- Eurozone banks finance dollar lending with flighty wholesale dollar liabilities, including U.S. money market fund funding.
- During the Eurozone sovereign crisis, U.S. MMFs sharply reduced funding to Eurozone banks.
- At the same time, the dollar-euro CIP basis widened, meaning that synthetic dollar funding through FX swaps became expensive.
- Eurozone banks reduced dollar lending relative to euro lending during the shock period.
- The effect is not observed in the same way for U.S. banks and is robust to borrower fixed effects.
- Borrowers previously dependent on Eurozone banks were less likely to obtain dollar loans and paid higher spreads if they did borrow.
- The effect is strongest for the banks most reliant on U.S. MMF funding.
- The paper’s main contribution is to show that currency-specific funding fragility can shape global bank lending and transmit shocks across borders.

8 Conclusion

8.1 Core conclusion

The paper shows that dollar funding shocks affect real credit supply. During the Eurozone crisis, Eurozone banks lost access to a major source of wholesale dollar funding and reduced dollar-denominated lending relative to euro-denominated lending. Borrowers tied to those banks faced lower access to dollar loans and higher spreads.

8.2 Economic intuition

The mechanism is simple but powerful. Eurozone banks fund euro lending with relatively sticky insured deposits, but fund dollar lending with more flighty wholesale liabilities. When bank credit risk rises, uninsured dollar creditors withdraw. Banks can try to replace direct dollar funding by borrowing euros and swapping into dollars, but if arbitrage capital is limited, CIP breaks down and the swap route becomes costly too. The bank therefore cuts dollar lending.

8.3 Incremental contribution revisited

The paper adds a currency dimension to the bank lending channel. The traditional bank-lending-channel story says weak banks reduce lending. This paper shows that weak global banks may reduce one currency of lending more than another, depending on how that currency is funded and whether swap markets can substitute across funding currencies.

8.4 Implications for global banking

The findings imply that the international use of the dollar creates fragility. When non-U.S. banks are major suppliers of dollar credit, shocks to their home-region credit quality or wholesale funding access can reduce dollar credit supply elsewhere. This can affect U.S. firms, global trade finance, and other dollar borrowers.

8.5 Policy implications

The analysis supports the logic of liquidity requirements for foreign banking organizations and helps explain the role of central bank dollar swap lines. Swap lines can reduce pressure on private FX swap markets by providing official dollar liquidity to foreign central banks, which can then on-lend dollars to local banks.

8.6 Limitations

The paper's event-study design is not a randomized experiment. Eurozone bank credit quality, sovereign stress, capital concerns, and dollar funding pressure all move together during 2011-2012. The paper addresses this by comparing currencies, banks, borrowers, and MMF-exposure groups, but some residual confounding is always possible.

8.7 Takeaway

Global banks do not simply move a single pool of funds across countries. Their lending depends on the currency structure of their assets and liabilities. When the dollar funding leg becomes fragile, dollar credit supply contracts, and firms dependent on those banks bear real costs.